

WHEN ARE EXACT CONSERVATION LAYERS PLUG-AND-PLAY?

IDENTIFYING TRAINING–INFERENCE PATH DEPENDENCE IN NEURAL PDE SURROGATES

Anonymous authors

Paper under double-blind review

ABSTRACT

Exact conservation layers are often treated as modular post-processors, but in autoregressive surrogates projection during training changes the learned representation and projection during inference changes future inputs. We separate these interventions with an eight-cell output-coordinate-by-training-by-inference cube and prove that projected training leaves the raw invariant channel gauge-nonidentifiable. In a prospectively frozen, 30-seed PDEBench Advection experiment, a small ordinary bundled interaction (-0.00189) conceals a three-way interaction of 0.22974 $[0.19644, 0.26345]$, material in $9/12$ mandatory resolution-horizon cells. Opposite-signed training and inference path credits cancel, and damage from removing projection appears only after autoregressive feedback. An independently frozen 30-seed 2D shallow-water replication gives $J = -86.69343$ $[-107.65055, -66.68345]$, material in $6/8$ cells, while projection jointly improves conservation, conserving RMSE, and positivity. A separately registered larger U-Net transfer gives $J = -0.08415$ $[-0.26665, 0.13334]$ is unresolved and only $3/12$ cells are material. Post-hoc replay places all eight primary U-Net cells below persistence, whereas the deployed residual FNO has forecast skill 0.9982 . A direct same-checkpoint input-gauge intervention yields an OOD feedback-to-baseline ratio of 8.68398 $[7.91007, 9.50538]$, above the frozen 0.10 threshold at all resolutions. Its seed-level association with final damage, like a local gradient association, remains unsupported. Exact enforcement can therefore become non-detachable: projected training can identify the deployed predictor while leaving its raw invariant component undefined in the tested FNOs. The U-Net result does not establish transfer among predictively useful architectures. Post-hoc, physically consistent Advection history shifts further separate projection dependence from learned shift equivariance.

1 INTRODUCTION

Learned simulators can achieve low average error while violating identities that the modeled system satisfies exactly. Architectural constraint layers, output projection, probabilistic conditioning, and physics penalties address this problem in climate emulation, PDE surrogate modeling, and neural operators (Beucler et al., 2021; Hansen et al., 2023; Harder et al., 2023; Liu et al., 2024). Exact enforcement has an unambiguous benefit: every returned state belongs to the admissible set. A different claim—that the constraint itself improved predictive accuracy or out-of-distribution (OOD) generalization—requires a controlled attribution design.

That control is easy to miss. Autoregressive simulators may predict an absolute next state $f_\theta(x_t)$ or an update $x_t + g_\theta(x_t)$. Exact conservation is naturally implemented by projecting the update into the invariant tangent space. Comparing this hard residual model only with an unconstrained absolute-output model changes two factors at once: output parameterization and enforcement. Residual updates are themselves a strong inductive bias in learned simulation (Pfaff et al., 2021). A favorable comparison therefore does not identify which factor deserves credit.

We study attribution rather than proposing another enforcement operator. The trained design crosses absolute/residual outputs with free/hard training; toggling projection on each locked checkpoint completes an eight-cell cube without optimization. This also exposes that projected training identifies raw outputs only modulo the invariant direction: a checkpoint may be accurate with its projector and unstable without it despite one-step conserving equivalence. Thus “plug-and-play” means safe same-checkpoint composition through rollout, not software compatibility or exact constraint satisfaction.

Our contributions are:

- an eight-cell intervention design that separately identifies output coordinate, training enforcement, inference enforcement, and their three-way path dependence;
- a projection-gauge non-identifiability result and a direct same-checkpoint intervention showing that an invariant-direction input change transfers materially into conserving dynamics;
- prospectively frozen 30-seed Advection and independently frozen 30-seed 2D shallow-water FNO evidence, plus a registered larger U-Net transfer that does not pass its gate; and
- orthogonal error and tangent-kernel analyses, including a frozen local mechanism test whose unsupported final association bounds rather than embellishes the explanation.

2 RELATED WORK

Encoding physical constraints. Hard constraints enforce analytic identities through architectures, invariant operators, projection, probabilistic correction, or learned correction (Beuclet et al., 2021; Richter-Powell et al., 2022; Liu et al., 2023; 2024; Hansen et al., 2023; Mouli et al., 2024; Zhang et al., 2023; Harder et al., 2023; Tanaka et al., 2025; Liu et al., 2025). Most directly, Duruisseau et al. (2024) compare an unconstrained FNO, post-hoc projection, constraint finetuning, and projected training; a contemporaneous submission ablates tangent-space projection and residual guidance (Anonymous Authors, 2026). These establish end-to-end feasibility and accuracy, but do not remove projection from the same constrained checkpoint, cross output coordinates, and follow that same-weight intervention through autoregressive feedback. Our target is this training-by-deployment interaction, not a replacement enforcement architecture.

Gradient interaction in physics learning. Composite PINN objectives can produce severe gradient-scale imbalance or directional conflict; adaptive scaling, dual-cone updates, and second-order preconditioning have been proposed to address these optimization pathologies (Yao et al., 2023; Hwang & Lim, 2024; Wang et al., 2025). Our mechanism question is narrower and structurally different: the two squared-error channels are orthogonal in output space, but a shared neural operator can map them to non-orthogonal parameter gradients. We do not propose a new optimizer. We use their alignment and an exact free-versus-hard one-step intervention to test whether this parameter-space coupling helps explain attribution.

Neural PDE evaluation. Fourier neural operators (FNOs) learn mappings between function spaces and support evaluation at unseen discretizations (Li et al., 2021). PDEBench provides standardized time-dependent PDE data, baselines, and metrics, including conservation-aware diagnostics (Takamoto et al., 2022a). Our experiments use its public Advection and two-dimensional shallow-water data, but change the evaluation question: rather than ranking a new operator, we estimate paired effects of parameterization and enforcement under the same locked weights.

3 A MATCHED ATTRIBUTION AUDIT

3.1 LINEAR INVARIANT AND ERROR CHANNELS

Let $x \in \mathbb{R}^n$ be the last observed state and y the true next state. For a nonzero $a \in \mathbb{R}^n$, the linear invariant is $a^\top y = a^\top x$. Define the orthogonal projectors

$$P = \frac{aa^\top}{\|a\|_2^2}, \quad Q = I - P. \quad (1)$$

For prediction $\hat{y}$ and error $e = \hat{y} - y$, Pe is the invariant-violating channel and Qe is the conserving channel. We report conserving RMSE, R_{cons} , from Qe . This metric deliberately does not reward a method merely for removing the mean error; total squared error decomposes as $\|e\|_2^2 = \|Qe\|_2^2 + \|Pe\|_2^2$.

The least-squares projection of an absolute prediction onto the affine invariant set is

$$\Pi_x(\hat{y}) = \hat{y} - P(\hat{y} - x). \quad (2)$$

Proposition 1 (Projection decoupling). *If $a^\top y = a^\top x$, then $\Pi_x(\hat{y}) - y = Q(\hat{y} - y)$. Consequently, projection removes the violating channel exactly and leaves the conserving-channel error unchanged.*

The proof is in Appendix A. In the experiment, the post-hoc projection of the trained free model is therefore an implementation control, not an independent learned baseline.

Proposition 2 (Projected-training gauge). *For any raw predictor $z(x)$ and scalar function $\alpha(x)$, $\Pi_x(z(x) + \alpha(x)a) = \Pi_x(z(x))$. Hence, in any hypothesis class closed under this transformation, a loss applied only after projection cannot identify the raw predictor’s invariant-channel component. Removing the projector from a hard-trained checkpoint selects an unidentified representative of the same deployed predictor.*

The proof is in Appendix B. Proposition 1 still makes the projected and unprojected predictions agree in conserving error for the same one-step input. For our FNO, the bias in its final pointwise map realizes at least every constant shift αa , so the closure condition is not merely an unrestricted-function abstraction. Autoregressive equivalence does not follow: an unprojected invariant-channel error changes the next input, where the network’s conserving response can differ. This yields a falsifiable signature of gauge exposure followed by state feedback—one-step equality followed by multi-step divergence.

Exact satisfaction of one invariant need not imply physical admissibility under another constraint. This distinction matters for conserved nonnegative fields such as water depth. Write $m(v) = n^{-1}\mathbf{1}^\top v$ for the uniform-grid mean.

Proposition 3 (Conservation–positivity compatibility). *For the uniform-mass invariant, $\Pi_x(z) = z + [m(x) - m(z)]\mathbf{1}$. If $z \geq 0$, the projected state is componentwise nonnegative if and only if $\min_i z_i \geq m(z) - m(x)$. Hence an orthogonal mass projection can enforce conservation exactly while introducing negative state values.*

This exact criterion motivates reporting invariant drift, conserving-channel forecast error, and admissible-set violations as three separate outcomes. In particular, clamping a projected field would silently compose two interventions and destroy the clean training-by-inference factorial.

The fixed-prediction statement also supplies a training-time null model. Let a coordinate choice $s \in \{A, R\}$ have base map $b_A(x) = 0$ or $b_R(x) = x$, and write the learned raw map as $z_\theta(x)$. Free and hard training have the same conserving prediction $Q(b_s(x) + z_\theta(x))$; they differ only because the free loss also contains the violating channel.

Proposition 4 (Channel-separable training null). *Suppose $\theta = (\theta_Q, \theta_P)$ and $z_\theta = z_Q(\theta_Q) + z_P(\theta_P)$, where $Pz_Q = 0$, $Qz_P = 0$, the regularizer separates over the two parameter blocks, and paired free and hard runs use identical minibatches, initial θ_Q , and a blockwise deterministic optimizer. Then their θ_Q iterates and conserving predictions are identical. Consequently, exact enforcement has zero conserving-channel training effect in that coordinate system.*

The proof is in Appendix D. A nonzero inference-matched training credit therefore rejects this separable-training null: training through the projector changed the admissible prediction through shared parameters, regularization, or optimization. A nonzero coordinate-by-training interaction makes the stronger statement that this coupling depends on absolute versus residual output coordinates. A bundled hard-versus-free rollout contrast is not enough for either conclusion because per-step projection also changes later inputs.

A local gradient diagnostic complements the training null. Shared parameters can couple the orthogonal output channels through the off-diagonal tangent-kernel block $\bar{Q}K_\theta\bar{P}$. Appendix E states the exact first-order transfer identity and a finite-step remainder bound. Because the actual optimizer is Adam, the empirical test also measures an exact matched one-Adam-step difference; neither diagnostic is assumed to explain the final rollout interaction.

3.2 MECHANISMS, PATH DEPENDENCE, AND PATH-AVERAGED CREDIT

The trained FNO body and parameter count are identical across mechanisms. With raw network output $g_\theta(x)$, the relevant arms are

$$\text{free} : \hat{y} = f_\theta(x), \quad (3)$$

$$\text{free_res} : \hat{y} = x + g_\theta(x), \quad (4)$$

$$\text{hard_abs} : \hat{y} = \Pi_x(f_\theta(x)), \quad (5)$$

$$\text{hard} : \hat{y} = x + Qg_\theta(x), \quad (6)$$

$$\text{projection} : \hat{y} = \Pi_x(f_\theta(x)). \quad (7)$$

For `hard_abs`, gradients pass through Π_x throughout training. For `projection`, θ is the already trained `free` parameter vector and projection is applied only as a derived evaluation transform; there is no fifth training run. The two confirmatory contrasts use conserving RMSE averaged over the same 30 seeds:

$$\Delta_{\text{param}} = R_{\text{cons}}(\text{free}) - R_{\text{cons}}(\text{free_res}), \quad \Delta_{\text{hard}} = R_{\text{cons}}(\text{hard}) - R_{\text{cons}}(\text{free_res}). \quad (8)$$

Positive Δ_{param} favors residual parameterization; negative Δ_{hard} favors hard enforcement. Together they give a path-specific decomposition of the bundled comparison, $R_{\text{cons}}(\text{free}) - R_{\text{cons}}(\text{hard}) = \Delta_{\text{param}} - \Delta_{\text{hard}}$. They do not estimate a fully crossed parameterization-by-training interaction.

Proposition 5 (Three-cell non-identification). *Without an additivity or other structural restriction on the missing potential outcome, observing only Y_{A0} , Y_{R0} , and Y_{R1} does not identify the factorial interaction or a unique path-averaged allocation. Infinitely many nonnegative values of the unobserved trained cell Y_{A1} preserve those three observations while giving distinct values of I_{path} and the two Shapley credits.*

The proof is in Appendix H. Consequently, neither a favorable three-arm sequence nor post-hoc projection of $A0$ can identify what training in the missing absolute-hard cell would have done. The projection is essential as an algebraic control, but it is not that trained intervention.

To identify that interaction, let $Y_{A0}, Y_{R0}, Y_{A1}, Y_{R1}$ denote paired-seed conserving RMSE for absolute versus residual coordinates (A, R) crossed with free versus hard enforcement $(0, 1)$. Define positive credits as error reductions,

$$P_0 = Y_{A0} - Y_{R0}, \quad P_1 = Y_{A1} - Y_{R1}, \quad (9)$$

$$E_A = Y_{A0} - Y_{A1}, \quad E_R = Y_{R0} - Y_{R1}. \quad (10)$$

The parameterization credit differs across enforcement levels by

$$I_{\text{path}} = P_0 - P_1 = E_A - E_R = Y_{A0} - Y_{R0} - Y_{A1} + Y_{R1}. \quad (11)$$

Thus I_{path} is simultaneously the factorial interaction and the discrepancy between two sequential credit-allocation paths. When it is not practically negligible, neither single path defines a unique contribution. We then use the exact two-factor Shapley averages

$$\phi_{\text{param}} = \frac{1}{2}(P_0 + P_1), \quad \phi_{\text{enforce}} = \frac{1}{2}(E_A + E_R), \quad \phi_{\text{param}} + \phi_{\text{enforce}} = Y_{A0} - Y_{R1}. \quad (12)$$

These are intervention credits computed from trained factorial cells, not feature-attribution scores. They average over the two possible orders instead of choosing the more favorable path after observing results (Shapley, 1953).

For the factorial primary cell, the smallest effect size of interest (SESOI) is $\delta = 0.1 \bar{Y}_{R0}$. The ordered frozen classification is: *material non-additivity* if the 95% paired-bootstrap interval for I_{path} lies wholly above $+\delta$ or below $-\delta$; *smaller statistical non-additivity* if that interval excludes zero but the first rule does not pass; *practical additivity* if the 90% interval lies wholly inside $[-\delta, +\delta]$; and *unresolved* otherwise. Thus both interaction and additivity are falsifiable, while an interval spanning both zero and a SESOI boundary receives no positive label. Seeds, not trajectories or rollout frames, are the inference units; each bootstrap draw resamples the complete paired four-cell seed vector. We use 50,000 draws. Twelve secondary interaction sign-flip tests receive Holm correction but cannot override the primary classification.

3.3 SEPARATING TRAINING ENFORCEMENT FROM INFERENCE PROJECTION

For multi-step rollouts, let Y_{cte} denote conserving RMSE in coordinate $c \in \{A, R\}$, with projector use during training $t \in \{0, 1\}$ and at every inference step $e \in \{0, 1\}$. The four trained models give $Y_{A00}, Y_{R00}, Y_{A11}, Y_{R11}$. Changing only the compatible forward map at evaluation supplies the other four cells; all such records take zero optimizer steps. Define the coordinate contrast $D_{te} = Y_{Ate} - Y_{Rte}$. Then

$$I_{\text{bundle}} = D_{00} - D_{11}, \quad (13)$$

$$T_0 = D_{00} - D_{10}, \quad T_1 = D_{01} - D_{11}, \quad (14)$$

$$E_0 = D_{00} - D_{01}, \quad E_1 = D_{10} - D_{11}, \quad (15)$$

$$J = T_0 - T_1 = E_0 - E_1 = D_{00} - D_{10} - D_{01} + D_{11}. \quad (16)$$

Here J is the coordinate-by-training-by-inference interaction. The path-averaged credits

$$\phi_{\text{train}} = \frac{1}{2}(T_0 + T_1), \quad \phi_{\text{infer}} = \frac{1}{2}(E_0 + E_1) \quad (17)$$

obey $\phi_{\text{train}} + \phi_{\text{infer}} = I_{\text{bundle}}$. We froze this cube while the fresh factorial was incomplete and no metric had been accessed. Its primary cell, bootstrap, SESOI, and ordered classification are inherited from the two-factor audit. The local gradient diagnostic is matched to T_1 , where both endpoints use projected inference; its association with I_{bundle} is retained only as a labeled secondary analysis.

4 PDEBENCH PROTOCOLS

Data and invariant. We use PDEBench 1D Advection with velocity $\beta = 0.4$ and periodic boundary conditions (Takamoto et al., 2022a;b). The native tensor contains 10,000 trajectories, 201 used frames, and 1024 spatial cells. The spatial mean (equivalently, uniform-grid mass) is invariant. A frozen admission gate checks file hashes, schema, finiteness, and target-mean drift. To construct 512- and 256-cell fields without changing the invariant, we average consecutive blocks of two and four native cells in float64 before casting to float32; this restriction preserves the discrete mean.

Trajectories 0–255 are the external test set, 1000–8999 form the training pool, and 9000–9255 were used only for outcome-blind data preflight. Each seed samples 2,048 unique training trajectories. Inputs are 10 frames and the target is the next frame at temporal stride five (physical time 0.05). Models train only on 256 cells and are evaluated without retraining on ID 256 and OOD 512 and 1024 cells at autoregressive horizons $\{1, 4, 16, 31\}$.

Model and optimization. The FNO has width 20, 12 Fourier modes, four spectral-plus-pointwise blocks, padding two, GELU activations, and projection width 128 (Li et al., 2021). Every trained arm uses Adam with learning rate 10^{-3} , weight decay 10^{-4} , batch size 50, 200 epochs, and a StepLR decay of 0.5 at epoch 100. There is no validation selection, early stopping, or hyperparameter search. Arms within a seed share the sampled trajectories, minibatch order, training windows, and initial parameter tensor. The formal factorial runs all four trained cells from identical initial tensors with fresh seeds 3000–3029; no earlier checkpoint, seed, or record is reused. Its protocol and analysis rule were frozen after a path-specific parent study but before seed 2999 preflight or any formal model run. The estimates below use only the declared factorial and transfer cohorts.

After observing the FNO result but before any U-Net record existed, we froze a transfer test that changes only the model body. The two-level fully convolutional 1D U-Net (Ronneberger et al., 2015) uses base width 20, GroupNorm, SiLU, average-pooling, transposed-convolution upsampling, skip connections, and spatial coordinates. Data, splits, training arms, optimization, evaluation cells, and decision rule are unchanged, with fresh paired seeds 7000–7029. Its fixed gate requires a material primary J and at least 6/12 material cells. This is neither a capacity-matched architecture effect (FNO: 23,937 parameters; U-Net: 66,941) nor a claim that a finite-receptive-field U-Net is a discretization-invariant operator. The post-hoc forecast-skill analysis below separately tests whether these checkpoints improve on persistence.

Before inspecting any factorial metric, we separately froze a mechanism audit that reconstructs each seed’s exact epoch-zero first minibatch. In absolute and residual coordinates it decomposes the full loss into $L_Q + L_P$, records $\langle g_Q, g_P \rangle$ and its cosine, and executes matched free and hard one-step

Adam updates from the identical initial tensor. Its local interaction is the absolute-minus-residual difference between post-step free-minus-hard conserving losses. The frozen primary test is the paired-seed Spearman association between that local interaction and the inference-matched training interaction T_1 , with a 50,000-draw 95% bootstrap interval. The original association with the bundled interaction remains secondary. A non-positive or zero-crossing interval does not support the proposed local mechanism.

After the cube was complete and that gradient association was unsupported, we froze a targeted same-checkpoint gauge intervention before any new record existed. From one ground-truth history it computes a hard checkpoint’s compatible raw state z_1 and projection p_1 , appends each to an otherwise identical history, and compares the conserving channel of the next raw prediction. The primary per-seed estimand is this feedback RMSE divided by the projected-history branch’s conserving RMSE, averaged across coordinates at OOD 512; materiality requires its 95% interval to lie wholly above 0.10. The exact 60-record run crosses 30 seeds and two coordinates with zero optimization. The numerical identity criterion uses a 10^{-6} absolute floor together with a worst-case IEEE-754 bound; non-invariant roundoff must remain below 0.1% of the projected baseline.

Prospective primary and integrity gates. The primary cell was fixed as OOD 512 at horizon 16; the other 11 cells were mandatory secondary results. The core contract is exactly 150 records: 30 seeds times four trained arms plus one zero-training projection. After its one-shot analyzer passed, all 120 trained checkpoints were hash-locked and used to derive the three missing cube cells per seed, exactly 90 additional records with zero optimization. The local gradient diagnostic and direct gauge intervention each add exactly 60 coordinate-by-seed records. Before each analyzer may run, it verifies record coverage, run-ID uniqueness, source and configuration hashes, seed pairing, parameter parity, finite metrics, checkpoint binding, the target invariant, and the horizon-one projection identity. The original confirmation intervals use the seed as inference unit; secondary sign-flip tests receive Holm correction across the 12 mandatory cells.

Independent shallow-water replication. Before downloading or inspecting its values, we separately froze PDEBench’s 2D radial-dam-break shallow-water task (version 8, 2D_rdb_NA_NA.h5). Its 1,000 trajectories contain 101 frames on a 128×128 non-periodic grid. Water depth supplies both a linear invariant (uniform-grid mass) and an independent admissibility outcome (nonnegativity). A single-field FNO2d with width 20 and eight modes per axis takes ten depth fields plus coordinates and predicts the next field; no normalization or clamp is applied. Each seed samples 512 trajectories from groups 200–999, trains at 64×64 for 100 epochs, and is evaluated on held-out groups 0–127 at ID-64 and native OOD-128, horizons $\{1, 4, 16, 31\}$. Seeds 6000–6029 use the same four trained arms and inference-toggle cube, giving exactly 150 core records, 90 zero-training derived records, and 120 locked checkpoints. OOD-128/horizon-16 is primary. OOD-128/horizon-31 prospectively tests whether projection’s mass gain trades against or improves negative-depth deficit, while reporting conserving RMSE as a third, dynamics-specific outcome.

5 RESULTS

5.1 THE FOUR TRAINED CELLS SHOW A SMALL BUNDLED INTERACTION

The fresh four-cell factorial first gives a plausible but incomplete story. In the primary OOD-512, horizon-16 cell, residual coordinates reduce conserving RMSE by 0.00154 [0.00080, 0.00231] under free training and by 0.00343 [0.00195, 0.00531] under hard training. Their difference, $I_{\text{bundle}} = -0.00189 [-0.00373, -0.00025]$, excludes zero but crosses the frozen SESOI boundary -0.00322 . Its ordered classification is therefore *statistical non-additivity below/crossing the SESOI*, not material non-additivity. The two-factor Shapley credits are both positive: $\phi_{\text{param}} = 0.00248 [0.00163, 0.00357]$ and $\phi_{\text{enforce}} = 0.00096 [0.00023, 0.00167]$. On these four deployed cells alone, both modeling choices appear useful and only mildly path dependent.

5.2 THE COMPLETE CUBE REVEALS PROJECTOR DEPENDENCE HIDDEN BY CANCELLATION

The missing inference cells reverse that interpretation (Figure 1). Applying the projector to free-trained checkpoints barely moves the primary means: $Y_{A00} = 0.03373$ versus $Y_{A01} = 0.03375$,

Table 1: Fresh-seed primary effects. Positive P_0 and P_1 favor residual coordinates. I_{bundle} is the four-trained-cell interaction; J is the full coordinate-by-training-by-inference interaction. Intervals are 95% paired bootstrap CIs.

Estimand	Mean	95% CI	Frozen interpretation
P_0	0.00154	[0.00080, 0.00231]	residual favored
P_1	0.00343	[0.00195, 0.00531]	residual favored
I_{bundle}	-0.00189	[-0.00375, -0.00025]	smaller non-additivity
J	0.22974	[0.19644, 0.26345]	material non-additivity
ϕ_{train}	0.11302	[0.09634, 0.12975]	countervailing credit
ϕ_{infer}	-0.11491	[-0.13166, -0.09818]	countervailing credit

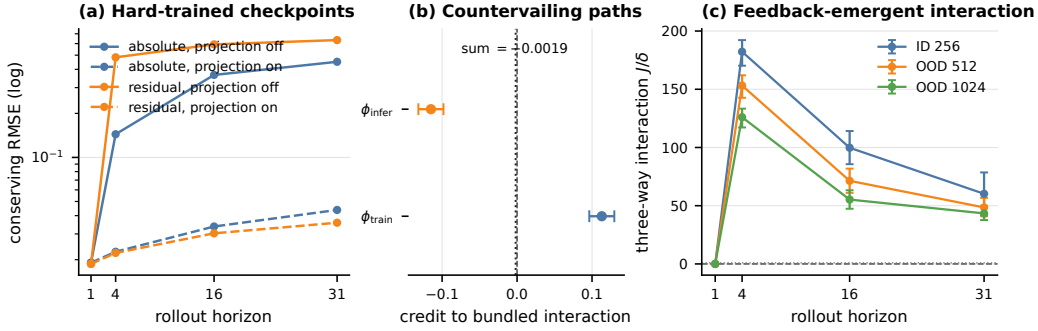


Figure 1: The projector becomes part of the learned autoregressive system. (a) At OOD 512, projected and unprojected hard-checkpoint evaluations agree in conserving RMSE at horizon one, as Proposition 1 requires, then diverge after feedback. Solid lines turn projection off; dashed lines retain it. (b) The primary path-averaged training and inference credits are large, opposite-signed, and sum to the small bundled interaction (vertical dotted line). (c) The three-way interaction, normalized by each cell’s SESOI, is near zero at horizon one and material from horizon four at every resolution. Error bars are 95% paired CIs.

and $Y_{R00} = 0.03219$ versus $Y_{R01} = 0.03218$. Removing it from hard-trained checkpoints instead raises conserving RMSE from $Y_{A11} = 0.03372$ to $Y_{A10} = 0.36627$, and from $Y_{R11} = 0.03029$ to $Y_{R10} = 0.59262$. These are evaluations of the same locked weights through compatible forward maps, with zero optimization.

Consequently, $J = 0.22974$ [0.19644, 0.26345] is 71.4 times the frozen SESOI and passes the material-nonadditivity rule; its Holm-adjusted sign-flip p -value is 0.00012. All 9 resolution cells at horizons 4, 16, and 31 are materially non-additive, whereas none of the three horizon-one cells is. The path-averaged contributions $\phi_{\text{train}} = 0.11302$ and $\phi_{\text{infer}} = -0.11491$ are individually large and opposite-signed, leaving only $I_{\text{bundle}} = -0.00189$ after cancellation. Thus the apparently benign deployed comparison does not imply modularity. Proposition 2 explains why hard training need not determine a usable raw invariant channel; the horizon-one identity followed by divergence in every registered horizon-four cell localizes the observed damage to autoregressive state feedback after that gauge is exposed.

5.3 THE REGISTERED U-NET TRANSFER DOES NOT PASS ITS GATE

Replacing the FNO body with the frozen 66,941-parameter U-Net changes the boundary. Its ordinary four-trained-cell interaction is nonzero, $I_{\text{bundle}} = 0.10234$ [0.00507, 0.19831], but completing the cube gives primary $J = -0.08415$ [-0.26665, 0.13334], classified *unresolved*. Only 3/12 mandatory cells are materially non-additive, versus the registered 6/12 requirement, so the architecture-transfer gate *did not pass*. All three material cells are ID-256 at horizons 4, 16, and 31; selecting them after the unresolved OOD-512 primary would reverse the prospective decision. Thus the FNO result is not architecture-general, and a nonzero four-cell interaction does not by itself establish training–inference path dependence.

Forecast skill bounds the transfer interpretation. Post-hoc replay evaluates all 240 Advection checkpoints on the same 256 test trajectories, using total-MSE skill $S = 1 - \text{MSE}/\text{MSE}_{\text{persistence}}$. At OOD-512, horizon 16, deployed residual FNO skill is 0.9982 [0.9979, 0.9986], whereas all eight U-Net cells have negative skill (Table 8). Intervals here jointly resample seeds and common trajectories. This rules out interpreting this U-Net result as a mechanism comparison between two predictively successful architectures; it does not isolate an intrinsic architecture effect from this fixed training recipe. Appendix N reports every resolution and horizon.

5.4 INDEPENDENT SHALLOW WATER REPRODUCES AND AMPLIFIES THE PATH DEPENDENCE

The prospectively frozen 2D shallow-water replication passes the same admission rule on a different equation, dimension, boundary geometry, and linear mass invariant. Its ordinary four-trained-cell interaction is 0.02954 [0.01534, 0.04369]. Completing the cube instead gives $J = -86.69343$ [-107.65055, -66.68345], whose magnitude is 5777.2 times the frozen SESOI (Holm $p = 8.000e-05$). The interaction is practically additive at horizon one at both resolutions, then material at horizons 4, 16, and 31, for 6/8 mandatory cells.

The diagnostic failure is sharply coordinate dependent: removing the projector from the hard-trained absolute checkpoint raises mean conserving RMSE from 0.09507 to 86.79537, whereas the corresponding residual checkpoint remains near 0.15006 versus 0.15006. The path-averaged training and inference credits are again large and opposing, -43.32405 and 43.35358, summing to the much smaller bundled interaction. This is not a claim that the unprojected hard model is a sensible deployment mode; it is evidence that the projector is constitutive of what projected training learned. Post-hoc paired-seed summaries show median absolute-coordinate removal harm 74.71 (IQR 42.71–128.71), positive in 28/30 seeds, showing broad seed-level harm within this fixed test cohort (Appendix M).

The independent admissibility test does not reveal a hidden cost. At OOD-128/horizon-31, adding projection to free-trained checkpoints improves maximum mass drift by 0.46656 [0.35540, 0.58242], conserving RMSE by 0.06994 [0.04168, 0.10034], and mean negative-depth deficit by $9.864e-05$ [$3.640e-08$, 0.00028]. Under the frozen sign convention and rule, this is a *resolved conservation-positivity synergy*, not a conservation–positivity trade-off.

5.5 A DIRECT INTERVENTION CONFIRMS GAUGE-TO-DYNAMICS FEEDBACK

The two-step same-checkpoint intervention isolates the proposed feedback link rather than inferring it from rollout timing. The first histories differ only by the raw prediction’s uniform invariant component; both then pass through the same compatible raw forward map. In the primary OOD-512 cell, the conserving-channel response has feedback RMSE 0.17447 [0.15881, 0.19064] and a feedback-to-projected-baseline ratio of 8.68398 [7.91007, 9.50538]. The latter is 86.8 times the frozen 0.10 practical threshold and is classified as *material gauge feedback*. All 3/3 registered resolutions reach the same classification, and both absolute and residual coordinates do so separately. The maximum numerical contamination is only $6.549e-05$ of the projected baseline, below the prespecified 0.00100 gate.

This establishes that an invariant-direction input perturbation can transfer directly into the learned admissible dynamics. It does not show that the two-step magnitude ranks eventual rollout failure across seeds: its Spearman association with coordinate-averaged horizon-16 removal harm is $\rho = 0.28765$ [-0.09679, 0.62675], classified *not supported*. We therefore claim existence and materiality of the feedback channel, not a seed-level quantitative explanation of every long-horizon divergence.

Physically consistent histories distinguish a second question. The gauge experiment compares two histories induced by projector removal. A post-hoc Advection analysis instead shifts all ten input frames by a common constant, or adds an analytically advecting sinusoid, and compares the next-step response with the corresponding physical response. For whole-history constant shifts, the OOD-512 normalized conserving residual is 0.0083–0.0106 across the four training arms. Hard-minus-free differences are positive in both coordinates (Table 11); exact output conservation therefore does not establish smaller constant-shift equivariance error. These reused checkpoints supply a complementary descriptive test, not another independent confirmation of the original gauge mechanism. Appendix O gives amplitudes, analytic references, and uncertainty.

Local gradient boundary. The frozen first-minibatch intervention finds an absolute-minus-residual one-Adam-step effect of 0.00015 [0.00012, 0.00019]. This rejects an exact local equality, but not the stronger mechanism test. Its paired-seed association with the inference-matched final training interaction $T_1 = -0.00186$ is only $\rho = 0.062$ with 95% bootstrap CI $[-0.297, 0.398]$, yielding the prespecified classification *not supported*. The analogous bundled association is also unsupported. We therefore do not claim that initial gradient alignment explains which seeds develop the final effect. The direct input-gauge intervention establishes a material feedback channel; the gradient diagnostic supplies a negative boundary on a competing local explanation.

Guarantee versus predictive attribution. Hard and projected rollouts satisfy the invariant while free models have no such guarantee. Because conserving RMSE excludes the mechanically corrected channel and fixed projection cannot change it (Proposition 1), our accuracy result concerns training and feedback through the projector, not a detachable post-hoc correction.

6 IMPLICATIONS AND CONCLUSION

Audit training and deployment jointly. A free absolute model versus a hard residual model estimates a bundled treatment; even four trained cells omit deployment dependence. Add and remove projection on both free- and hard-trained checkpoints while retaining weights, because retraining changes the intervention. The usual asymmetric comparison—adding projection to a free checkpoint without removing it from a hard one—cannot detect learned projector dependence.

Do not interpret raw outputs of a projected model as identified predictions. Because the hard loss is invariant along the projection gauge, raw hard-checkpoint values are diagnostic latent quantities, not trained standalone forecasts. Model surgery, distillation, quantization, or fallbacks that bypass a “post-processing” layer can therefore change dynamics.

Report orthogonal channels and failed mechanisms. Report both $\|Qe\|$ and $\|Pe\|$: total RMSE rewards mechanically deleted drift, whereas conservation error alone hides admissible-direction damage. Nonlinear conditions such as positivity remain separate (Proposition 3). The direct gauge response is material, but neither its final-harm association nor the local-gradient association is supported. U-Net performance below persistence also prevents a mechanism claim spanning predictively useful architectures. Exact conservation and physical-history equivariance are distinct properties. Exact enforcement should therefore be evaluated as part of the trained transition, not as a detachable correction.

7 LIMITATIONS AND NON-CLAIMS

Evidence spans two public datasets, one coefficient/scenario per equation, uniform mass, and resolution shifts plus post-hoc Advection history interventions; it does not cover nonlinear invariants, generic noise, coefficient shifts, or broad initial-condition distributions. Both positive PDE transfers use FNOs. The larger U-Net does not pass its frozen gate, performs below persistence in the primary cell, and is not capacity matched, so architecture scope remains open. The shallow-water surrogate observes depth but not momentum or velocity, making its evidence partial-state rather than full-system fidelity. Projector removal is diagnostic, and gauge non-identification does not force every optimizer to select an unstable representative. Credits, the training null, and the 10% SESOI condition on the fixed design; cells share trained models, and intervals condition on the frozen trajectories. The post-hoc crossed bootstrap additionally resamples those observed trajectories; it does not establish coverage for every initial-condition distribution. Constant-shift and advecting-sine symmetries apply here to linear periodic Advection, not to shallow-water depth.

REPRODUCIBILITY STATEMENT

Appendix J records the data version, split, conservative restriction, model, optimization, seeds, inference units, and frozen decision rules. The supplementary artifact contains the Hydra configurations, immutable data-file and checkpoint locks, 450 core records, 270 zero-training cube records, 60 local-gradient records, 60 direct-gauge records, validation tests, one-shot analyzers, and the exact JSON

used to generate every reported empirical estimate and Figure 1. Checkpoint-lock manifests bind all 360 final weight files by path, byte count, payload identity, and SHA-256; this does not imply that the weight bytes are embedded in the PDF supplement. All formal records bind to source and protocol hashes. The accompanying revision data view contains all 192 checkpoint-skill cells and 144 physical-history response cells, both interval definitions, and generated tables. Their post-hoc status remains separate from the original confirmation. Full checkpoints and trajectory arrays are retained in the recovery archive.

AI USE STATEMENT

Generative AI tools were used to help formulate and critically revise the attribution framework, mathematical statements and proof drafts, hypotheses and the prospective experiment protocol; implement and test experiment and analysis code; prepare the conservative data interface; interpret frozen outputs; discover and summarize related literature; generate figure code; format references; and draft and edit the manuscript. They were not used to generate the underlying PDEBench observations. The authors reviewed the primary sources, executable tests, immutable hashes, analysis outputs, mathematical claims, and all AI-assisted text and code. The authors take responsibility for the final content, including claims and artifacts produced with AI assistance.

REFERENCES

- Anonymous Authors. Flow-based automatic neural operator with hard physical constraints. Public submission to the International Conference on Learning Representations, 2026. URL <https://openreview.net/forum?id=IRGAMx3f6N>. OpenReview submission; accessed 2026-09-01.
- Tom Beucler, Michael Pritchard, Stephan Rasp, Jordan Ott, Pierre Baldi, and Pierre Gentine. Enforcing analytic constraints in neural networks emulating physical systems. *Physical Review Letters*, 126(9):098302, 2021. doi: 10.1103/PhysRevLett.126.098302.
- Valentin Duruisseaux, Miguel Liu-Schiaffini, Julius Berner, and Anima Anandkumar. Towards enforcing hard physics constraints in operator learning frameworks. In *ICML Workshop on AI for Science*, 2024. URL <https://openreview.net/forum?id=Zvxxm14Rd1F>.
- Derek Hansen, Danielle C. Maddix, Shima Alizadeh, Gaurav Gupta, and Michael W. Mahoney. Learning physical models that can respect conservation laws. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 12469–12510, 2023.
- Paula Harder, Alex Hernandez-Garcia, Venkatesh Ramesh, Qidong Yang, Prasanna Sattigeri, Daniela Szwarcman, Campbell D. Watson, and David Rolnick. Hard-constrained deep learning for climate downscaling. *Journal of Machine Learning Research*, 24(365):1–40, 2023.
- Youngsik Hwang and Dong-Young Lim. Dual cone gradient descent for training physics-informed neural networks. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-3128.
- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- Chaoyu Liu, Yangming Li, Zhongying Deng, Chris Budd, and Carola-Bibiane Schönlieb. Conservation-preserved fourier neural operator through adaptive correction. *arXiv preprint arXiv:2505.24579*, 2025.
- Ning Liu, Yue Yu, Huaqian You, and Neeraj Tatikola. INO: Invariant neural operators for learning complex physical systems with momentum conservation. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 6822–6838, 2023.

- Ning Liu, Yiming Fan, Xianyi Zeng, Milan Klöwer, Lu Zhang, and Yue Yu. Harnessing the power of neural operators with automatically encoded conservation laws. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 30965–30997, 2024.
- S. Chandra Mouli, Danielle C. Maddix, Shima Alizadeh, Gaurav Gupta, Andrew Stuart, Michael W. Mahoney, and Bernie Wang. Using uncertainty quantification to characterize and improve out-of-domain learning for PDEs. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 36372–36418, 2024.
- Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter W. Battaglia. Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations*, 2021.
- Jack Richter-Powell, Yaron Lipman, and Ricky T. Q. Chen. Neural conservation laws: A divergence-free perspective. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pp. 234–241. Springer, 2015. doi: 10.1007/978-3-319-24574-4_28.
- Lloyd S. Shapley. A value for n -person games. In Harold W. Kuhn and Albert W. Tucker (eds.), *Contributions to the Theory of Games II*, pp. 307–317. Princeton University Press, 1953.
- Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 35, 2022a.
- Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. PDEBench data, version 8. DaRUS, 2022b.
- Yusuke Tanaka, Takaharu Yaguchi, Tomoharu Iwata, and Naonori Ueda. Energy-consistent neural operators for hamiltonian and dissipative partial differential equations. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pp. 1882–1890, 2025.
- Sifan Wang, Ananyae Kumar Bhartari, Bowen Li, and Paris Perdikaris. Gradient alignment in physics-informed neural networks: A second-order optimization perspective. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-5615.
- Jiachen Yao, Chang Su, Zhongkai Hao, Songming Liu, Hang Su, and Jun Zhu. MultiAdam: Parameter-wise scale-invariant optimizer for multiscale training of physics-informed neural networks. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 39702–39721, 2023.
- Wang Zhang, Tsui-Wei Weng, Subhro Das, Alexandre Megretski, Luca Daniel, and Lam M. Nguyen. ConCerNet: A contrastive learning based framework for automated conservation law discovery and trustworthy dynamical system prediction. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 41694–41714, 2023.

A PROOF OF PROJECTION DECOUPLING

Because the target satisfies the invariant, $P(y - x) = 0$. Therefore

$$\Pi_x(\hat{y}) - y = \hat{y} - P(\hat{y} - x) - y \quad (18)$$

$$= e - P((\hat{y} - y) + (y - x)) \quad (19)$$

$$= e - Pe = Qe. \quad (20)$$

Applying P gives zero and applying Q returns Qe . Orthogonality of P and Q gives the squared-error decomposition.

B PROOF OF THE PROJECTED-TRAINING GAUGE

Because $Pa = a$ for the rank-one projector onto $\text{span}(a)$,

$$\Pi_x(z + \alpha a) = z + \alpha a - P(z + \alpha a - x) \quad (21)$$

$$= z - P(z - x) + \alpha(a - Pa) \quad (22)$$

$$= \Pi_x(z). \quad (23)$$

Thus every realizable raw map in the equivalence class $\{z + \alpha a : \alpha : \mathcal{X} \rightarrow \mathbb{R}\}$ induces exactly the same projected prediction and loss. Projected observations cannot distinguish members of this class, so the raw invariant-channel representative is unidentified without an additional restriction. This is an output-space statement; shared parameters may still couple the identified and unidentified channels, as analyzed by Propositions 4 and 6.

C PROOF OF CONSERVATION-POSITIVITY COMPATIBILITY

For $a = \mathbf{1}$, the projector subtracts the prediction’s mean excess: $\Pi_x(z)_i = z_i + m(x) - m(z)$. Therefore $\Pi_x(z)_i \geq 0$ for every coordinate exactly when $z_i \geq m(z) - m(x)$ for every coordinate, which is equivalent to $\min_i z_i \geq m(z) - m(x)$. Taking any nonnegative z with $0 \leq \min_i z_i < m(z) - m(x)$ proves that exact mass projection can create a negative component.

D PROOF OF THE CHANNEL-SEPARABLE TRAINING NULL

For either coordinate choice, the conserving component of both the free and hard prediction is $Q(b_s + z_Q(\theta_Q))$ because $Qz_P = 0$. The free squared loss decomposes orthogonally into a term depending only on θ_Q and a violating-channel term depending only on θ_P ; the hard loss retains the first term and makes the second constant. The separable regularizer preserves this equality of the θ_Q objectives. Thus paired minibatches give identical gradients for θ_Q at every step. Equal initialization and a blockwise deterministic optimizer imply by induction that all θ_Q iterates, and hence all conserving predictions, are identical. The paired conserving-error difference is therefore zero.

E LOCAL CHANNEL-TRANSFER STATEMENTS

The first optimization step yields a sharper local diagnostic. Decompose the free squared loss as $L = L_Q + L_P$ and write $g_Q = \nabla L_Q(\theta)$ and $g_P = \nabla L_P(\theta)$.

For a stacked minibatch prediction h_θ , let $e = h_\theta - y$, let J_θ be its parameter Jacobian, and let $\bar{P}$ and $\bar{Q}$ apply P and Q independently to every example. The finite-width empirical tangent kernel is $K_\theta = J_\theta J_\theta^\top$ (Jacot et al., 2018). Take $L_Q = \frac{1}{2}\|\bar{Q}e\|^2$ and $L_P = \frac{1}{2}\|\bar{P}e\|^2$; a common normalization only rescales the step size.

Proposition 6 (Tangent-kernel channel transfer). *Let $h_\theta^{\text{lin}}(\theta + u) = h_\theta + J_\theta u$. From the same parameters, take a free gradient step $\theta_F^+ = \theta - \eta(g_Q + g_P)$ and a hard step $\theta_H^+ = \theta - \eta g_Q$. Then*

$$\bar{Q}[h_\theta^{\text{lin}}(\theta_F^+) - h_\theta^{\text{lin}}(\theta_H^+)] = -\eta \bar{Q} K_\theta \bar{P} e, \quad (24)$$

$$\langle g_Q, g_P \rangle = (\bar{Q}e)^\top K_\theta (\bar{P}e). \quad (25)$$

Thus the violating-channel gradient has zero first-order effect on the conserving prediction for every violating residual if and only if the off-diagonal channel block $\bar{Q} K_\theta \bar{P}$ vanishes.

The proof is in Appendix F. This identity links the global separable-training null to a measurable local mechanism. Shared parameters need not produce a nonzero off-diagonal block, but any nonzero block is a route by which hard enforcement can change admissible predictions even though P and Q are orthogonal in output space.

Proposition 7 (One-step gradient coupling). *If ∇L_Q is β -Lipschitz, then a gradient-descent step of size η obeys*

$$L_Q(\theta - \eta(g_Q + g_P)) - L_Q(\theta - \eta g_Q) = -\eta \langle g_Q, g_P \rangle + r, \quad (26)$$

where $|r| \leq \frac{\beta \eta^2}{2} (\|g_Q + g_P\|_2^2 + \|g_Q\|_2^2)$.

The proof is in Appendix G. Negative gradient alignment therefore makes removing the violating-channel gradient locally favorable for L_Q , up to the stated remainder; positive alignment reverses that direction. Output coordinates can change g_P , providing a testable local mechanism for the factorial interaction. Because the actual optimizer is Adam and training is nonlinear, we prospectively record both the gradient geometry and the exact matched one-Adam-step difference; neither is assumed to explain the final rollout interaction. This use of gradient alignment follows the broader composite-loss diagnostic literature (Hwang & Lim, 2024; Wang et al., 2025), while the orthogonal conservation channels and matched enforcement intervention are specific to our audit.

F PROOF OF TANGENT-KERNEL CHANNEL TRANSFER

The two parameter displacements differ by $\theta_F^+ - \theta_H^+ = -\eta g_P$. Because $g_Q = J_\theta^\top \bar{Q}e$ and $g_P = J_\theta^\top \bar{P}e$, the linearized conserving-output difference is

$$\bar{Q}J_\theta(\theta_F^+ - \theta_H^+) = -\eta \bar{Q}J_\theta J_\theta^\top \bar{P}e = -\eta \bar{Q}K_\theta \bar{P}e. \quad (27)$$

The inner-product identity follows from $g_Q^\top g_P = (\bar{Q}e)^\top J_\theta J_\theta^\top (\bar{P}e)$. If $\bar{Q}K_\theta \bar{P} = 0$, the first expression vanishes for every violating residual. Conversely, vanishing for every vector in the range of $\bar{P}$ implies $\bar{Q}K_\theta \bar{P} = 0$.

G PROOF OF THE ONE-STEP GRADIENT-COUPPLING BOUND

For a function with β -Lipschitz gradient, the descent lemma gives $L_Q(\theta + u) = L_Q(\theta) + g_Q^\top u + r_u$ with $|r_u| \leq \beta \|u\|_2^2/2$. Apply it to $u = -\eta(g_Q + g_P)$ and $v = -\eta g_Q$, then subtract the two expansions. Their linear terms differ by $-\eta g_Q^\top g_P$ and the triangle inequality gives

$$|r_u - r_v| \leq \frac{\beta \eta^2}{2} (\|g_Q + g_P\|_2^2 + \|g_Q\|_2^2), \quad (28)$$

which is Equation 26.

H PROOF OF THREE-CELL NON-IDENTIFICATION

Fix arbitrary nonnegative observed values Y_{A0} , Y_{R0} , and Y_{R1} , and let $C = Y_{A0} - Y_{R0} + Y_{R1}$. For any $c \leq C$, choose

$$Y_{A1} = Y_{A0} - Y_{R0} + Y_{R1} - c. \quad (29)$$

This value is nonnegative, and substitution into Equation 11 gives $I_{\text{path}} = c$ while leaving every observed cell unchanged. The half-line $c \leq C$ contains infinitely many choices. Because both $P_1 = Y_{A1} - Y_{R1}$ and $E_A = Y_{A0} - Y_{A1}$ depend on this unobserved value, the Shapley credits in Equation 12 are also unidentified. Proposition 1 constrains the conserving error of a fixed projected prediction, not the parameters learned by training through the hard absolute-output layer, so projection cannot supply the missing intervention.

I COMPLETE FROZEN INTERVENTION-CUBE SUMMARY

Table 2 lists the three-way interaction in every mandatory resolution-horizon cell. It is not a set of twelve replications: all rows reuse the same paired checkpoints, and Holm correction is applied to their sign-flip tests. The frozen ordered classification uses each cell’s own $\delta = 0.1\bar{Y}_{R00}$.

Table 2: All mandatory PDEBench Advection intervention-cube cells. J/δ gives effect size relative to the prespecified practical threshold; p is Holm-adjusted.

Case	Horizon	J	95% CI	J/δ	Class	Holm p
ID 256	1	2.52×10^{-10}	$[4.29 \times 10^{-11}, 4.76 \times 10^{-10}]$	0.0	smaller	0.0932
ID 256	4	0.3256	[0.3041, 0.3434]	182.2	material	0.0001
ID 256	16	0.2205	[0.1893, 0.2522]	99.8	material	0.0001
ID 256	31	0.1667	[0.1162, 0.2175]	60.2	material	0.0001
OOD 512	1	1.10×10^{-10}	$[-1.96 \times 10^{-10}, 4.10 \times 10^{-10}]$	0.0	additive	0.7632
OOD 512	4	0.3397	[0.3164, 0.3593]	153.2	material	0.0001
OOD 512	16	0.2297	[0.1964, 0.2634]	71.4	material	0.0001
OOD 512	31	0.1914	[0.1612, 0.2237]	48.5	material	0.0001
OOD 1024	1	-1.24×10^{-10}	$[-3.89 \times 10^{-10}, 1.48 \times 10^{-10}]$	0.0	additive	0.7632
OOD 1024	4	0.3450	[0.3209, 0.3651]	126.0	material	0.0001
OOD 1024	16	0.2338	[0.2004, 0.2676]	55.3	material	0.0001
OOD 1024	31	0.2203	[0.1913, 0.2531]	43.4	material	0.0001

J PRIMARY EIGHT CELLS AND ARTIFACT IDENTITY

Table 3 makes the full primary intervention transparent. Training and inference projection are separate binary factors. The four off-diagonal inference cells reuse locked weights and take zero optimizer steps.

Table 3: Mean conserving RMSE in all eight OOD-512, horizon-16 cells across 30 paired seeds.

	Training	Inference	Absolute	Residual
free		projector off	0.03373	0.03219
free		projector on	0.03375	0.03218
hard		projector off	0.36627	0.59262
hard		projector on	0.03372	0.03029

For reference, the primary conserving-RMSE means are 0.03373 for `free`, 0.03219 for `free_res`, 0.03372 for `hard_abs`, 0.03029 for `hard`, and 0.03375 for free-trained projection. These descriptive quantities do not define extra hypothesis tests. A projection rollout feeds corrected states back into the FNO, so after its first step its inputs differ from an unprojected rollout; the fixed-input identity is checked at horizon one rather than assumed at horizon 16.

The dataset is DaRUS PDEBench version 8, file ID 255674, 1D_Advection_Sols_beta0.4.hdf5, released under CC BY 4.0. The observed file size is 8,232,966,952 bytes. Its content hashes are MD5: d595bbfd2c659df995a93cd40d6ea568 and SHA256: d973ff2bb3c2a5edf42957ff029b78672451532ebd6bf475848ff23cfd3ee3b6. The fresh core JSONL/analysis SHA-256 prefixes are d0c8257af992/15ab5898699e; the zero-training cube prefixes are bee99e1b15d0/d12dc6d088d2; and the local-diagnostic prefixes are fd5251a63f6d/e6edc0803477. The direct-gauge data/analysis prefixes are 285ba66aeb84/87ed9676386e. The supplementary manifest records and verifies the complete hashes rather than these display prefixes.

The original confirmation uses seeds as its inference unit; the post-hoc skill and history analyses also report crossed seed/trajectory intervals. All mechanisms within a seed use the same training subset, deterministic window cycle, minibatch order, initialization tensor, and evaluation trajectories. No model outcome was read before complete coverage and the integrity gates passed.

K REGISTERED U-NET ARCHITECTURE-TRANSFER BOUNDARY

Table 4 reports the complete mandatory transfer grid rather than only the three favorable ID rows. The gate required both a material OOD-512/horizon-16 primary and at least 6/12 material cells. The primary is *unresolved*, and only 3/12 cells are material, so the registered transfer *did not pass*. Six cells show smaller statistical non-additivity, one is practically additive, and two are unresolved. This pattern is a boundary result, not evidence of absence in every U-Net cell.

Table 4: All mandatory U-Net intervention-cube cells. J/δ uses the same prespecified practical threshold as the FNO audit; p is Holm-adjusted.

Case	Horizon	J	95% CI	J/δ	Class	Holm p
ID 256	1	2.21×10^{-10}	$[8.98 \times 10^{-12}, 4.26 \times 10^{-10}]$	0.0	smaller	0.2483
ID 256	4	-0.0791	$[-0.1135, -0.0428]$	-27.3	material	0.0018
ID 256	16	0.4236	$[0.2522, 0.6406]$	26.5	material	0.0002
ID 256	31	0.8233	$[0.5034, 1.2144]$	23.1	material	0.0001
OOD 512	1	-5.99×10^{-10}	$[-1.15 \times 10^{-9}, -3.44 \times 10^{-11}]$	0.0	smaller	0.2483
OOD 512	4	-0.0469	$[-0.0619, -0.0322]$	-1.0	smaller	0.0001
OOD 512	16	-0.0841	$[-0.2666, 0.1333]$	-0.6	unresolved	1.0000
OOD 512	31	-0.0188	$[-0.3469, 0.3441]$	-0.1	unresolved	1.0000
OOD 1024	1	2.86×10^{-10}	$[-4.69 \times 10^{-10}, 1.08 \times 10^{-9}]$	0.0	additive	1.0000
OOD 1024	4	-0.0537	$[-0.0723, -0.0342]$	-0.8	smaller	0.0002
OOD 1024	16	-0.3506	$[-0.5808, -0.0727]$	-1.6	smaller	0.0747
OOD 1024	31	-0.6128	$[-1.1199, -0.0811]$	-1.7	smaller	0.1933

Table 5: Mean U-Net conserving RMSE in the eight primary OOD-512, horizon-16 cells across 30 paired seeds. Off-diagonal inference cells reuse locked weights and take zero optimizer steps.

	Training	Inference	Absolute	Residual
free		projector off	0.86332	1.47514
free		projector on	0.86423	1.47043
hard		projector off	0.82239	1.45803
hard		projector on	0.83205	1.54622

The U-Net merged core/analysis SHA-256 prefixes are ae0dda1c95e5/6e3252b5a1d3; the 120-checkpoint lock is e7d43d3d3ce9; and the cube/analysis prefixes are 01a23a0b77ff/0e2ad4bdf6f9. The formal seeds are 7000–7029. Both fixed 15-seed workers exited before the core analyzer ran; the cube followed only after exact 150-record coverage and checkpoint-lock verification. The supplementary manifest records the full hashes.

L INDEPENDENT SHALLOW-WATER ARTIFACT IDENTITY

Table 6: All eight shallow-water primary OOD-128, horizon-16 cells. Entries are mean conserving RMSE across 30 paired seeds; off-diagonal cells change only the inference forward map of a locked checkpoint and take zero optimizer steps.

	Training	Inference	Absolute	Residual
free		projector off	0.12461	0.15006
free		projector on	0.11774	0.15006
hard		projector off	86.79537	0.15006
hard		projector on	0.09507	0.15006

The replication uses DaRUS PDEBench version 8, file ID 133021, 2D_rdb_NA_NA.h5, with 6,626,098,972 bytes and hashes MD5:75d838c47aa410694bdc912ea7f22282 and SHA256:

28f0c33723d70eebb420fc170e94b675c18e032fb697dcef080e114ca9645e3a. The data gate verifies all 1,000 group names, tensor and coordinate schemas, finiteness, strict target positivity, mass drift, and both block-average restrictions. The merged core JSONL/analysis SHA-256 prefixes are 5e486167a2a8/d194df9d3f8d; the checkpoint lock prefix is 772c52f9dbaa; and the zero-training cube/analysis prefixes are 69dfa936fd83/f8492362b377. Exact hashes are retained in the supplementary manifest.

M POST-HOC PAIRED-SEED REMOVAL-HARM DISTRIBUTIONS

For each original checkpoint pair, removal harm is conserving RMSE without minus with the projector at the primary OOD resolution and horizon 16. Positive values indicate harm. These post-hoc summaries reuse the original records, preserve all 30 seeds, and condition on their common test trajectories. The bootstrap mean intervals below are descriptive and do not replace the original frozen interaction intervals or SESOI. Median, interquartile range and sign counts show the empirical seed distribution; they do not constitute another confirmatory test.

Table 7: Conserving-RMSE removal harm in each dataset’s field units. Magnitudes cannot be compared directly between Advection and shallow water. The complete mean/median/IQR/min/max summaries for all eight cells, both RMSE metrics, and all resolutions/horizons are in the accompanying scientific data view.

Cohort / coordinate	Mean [95% CI]	Median [IQR]	Positive seeds
Advection FNO / A	0.333 [0.299, 0.365]	0.314 [0.259, 0.433]	30/30
Advection FNO / R	0.562 [0.557, 0.569]	0.558 [0.553, 0.570]	30/30
Advection U-Net / A	-0.010 [-0.040, 0.021]	-0.010 [-0.067, 0.059]	13/30
Advection U-Net / R	-0.088 [-0.274, 0.137]	-0.274 [-0.429, 0.096]	8/30
Shallow-water FNO / A	86.700 [66.517, 107.565]	74.712 [42.707, 128.710]	28/30
Shallow-water FNO / R	2.11e-08 [1.80e-10, 4.28e-08]	0.000 [-5.33e-09, 1.27e-08]	9/30

N POST-HOC FORECAST SKILL ON THE LOCKED ADVECTION CHECKPOINTS

The 120 FNO and 120 U-Net checkpoints comprise four trained arms and 30 paired seeds per architecture. Two inference policies give all eight cells on the existing 256 shared trajectories. We compute aggregate total MSE across seeds, trajectories, and spatial cells before dividing by persistence MSE; this is not an average of per-trajectory skill ratios. Persistence repeats the last observed field, and the analytic control translates it at the known Advection velocity. This analysis reuses both weights and test trajectories and remains post-hoc.

The 10,000-draw crossed bootstrap independently resamples seeds and shared trajectories while preserving pairing across cells. Seed-only intervals condition on those trajectories; both sets are in the scientific JSON. Table 8 reports descriptive 95% crossed intervals for every primary cell. For the two architecture-specific R11 summaries, the separately adjusted intervals use 97.5% coverage; all other cell intervals are marginal, not simultaneous. A negative skill means that total forecast MSE exceeds persistence MSE. In particular, very negative A10 and R10 FNO skill measures raw hard-checkpoint removal damage, not deployed accuracy.

Table 8: Post-hoc total-MSE skill at OOD-512/horizon-16. The original conserving-RMSE cube and its registered gate remain unchanged. A/R denote absolute/residual outputs; the two digits indicate training and inference projection.

Cell	FNO skill [95% CI]	U-Net skill [95% CI]
A00	0.9973 [0.9968, 0.9978]	-0.4446 [-0.6119, -0.2976]
A01	0.9978 [0.9973, 0.9982]	-0.4309 [-0.5930, -0.2885]
A10	-280.06 [-479.01, -115.64]	-1.2798 [-1.6019, -1.0112]
A11	0.9978 [0.9973, 0.9982]	-0.3238 [-0.4650, -0.1976]
R00	0.9975 [0.9970, 0.9979]	-3.2884 [-4.0935, -2.6208]
R01	0.9980 [0.9976, 0.9984]	-3.2580 [-4.0574, -2.5961]
R10	-265.31 [-311.56, -228.54]	-58.45 [-79.68, -40.51]
R11	0.9982 [0.9979, 0.9986]	-3.7688 [-4.7886, -2.9304]

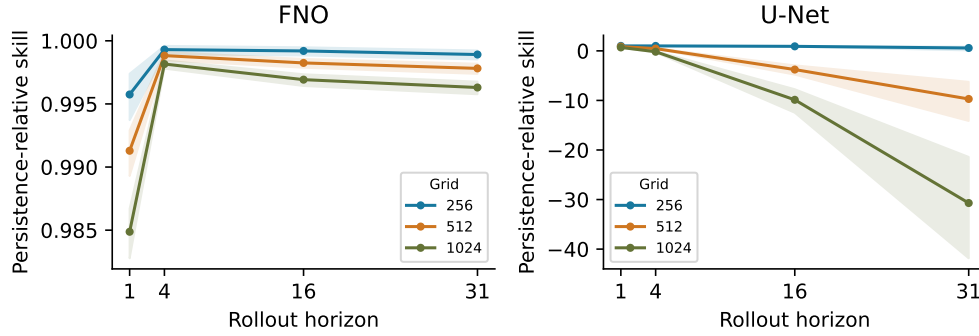


Figure 2: Deployed hard residual (R11) skill across all mandatory Advection grids and horizons. Shading gives pointwise 95% crossed intervals. Axes differ between architectures. This is post-hoc replay; the complete eight-cell estimates are in the accompanying CSV and JSON.

Table 9: All mandatory FNO cells for the four trained/deployed arms: total-MSE skill.

Grid/horizon	A00	A11	R00	R11
256/1	0.9930	0.9949	0.9938	0.9957
256/4	0.9991	0.9992	0.9992	0.9993
256/16	0.9985	0.9989	0.9988	0.9992
256/31	0.9972	0.9980	0.9980	0.9989
512/1	0.9906	0.9909	0.9913	0.9913
512/4	0.9987	0.9988	0.9988	0.9988
512/16	0.9973	0.9978	0.9975	0.9982
512/31	0.9952	0.9966	0.9959	0.9978
1024/1	0.9869	0.9852	0.9874	0.9849
1024/4	0.9981	0.9981	0.9981	0.9982
1024/16	0.9957	0.9964	0.9958	0.9969
1024/31	0.9926	0.9947	0.9934	0.9963

Table 10: Corresponding U-Net skill. This fixed recipe does not establish transfer among predictively useful architectures.

Grid/horizon	A00	A11	R00	R11
256/1	0.9954	0.9990	0.9993	0.9992
256/4	0.9948	0.9985	0.9974	0.9967
256/16	0.9540	0.9858	0.9340	0.9135
256/31	0.8735	0.9613	0.6895	0.5794
512/1	0.8055	0.8153	0.8669	0.8624
512/4	0.6545	0.6609	0.4826	0.4582
512/16	-0.4446	-0.3238	-3.2884	-3.7688
512/31	-0.1711	0.0482	-7.5920	-9.7065
1024/1	0.5747	0.5809	0.7058	0.6936
1024/4	0.3022	0.3074	-0.1280	-0.1759
1024/16	-0.3969	-0.2059	-8.4407	-9.8754
1024/31	-0.4227	0.0019	-26.39	-30.70

O POST-HOC PHYSICAL-HISTORY INTERVENTIONS

For linear periodic Advection, a constant shift of every history frame is a physical symmetry: the next field shifts by the same constant, so its conserving (Q) response is zero. For a sinusoid, the reference instead transports that perturbation at velocity $\beta = 0.4$ through both the ten history times and the next prediction time. Its analytic Q response is nonzero. For each existing FNO checkpoint we compare the deployed next-step responses with and without these interventions. We divide the spatial RMS of the response minus its analytic Q reference by $|\alpha|$, using amplitudes $\alpha/\sigma \in \{-0.5, -0.1, 0.1, 0.5\}$. The scale $\sigma = 0.7960596902$ is the standard deviation of native-grid values in the first ten frames of seed 3000’s 2,048 training trajectories; test values do not determine it.

All 120 checkpoints use the same 256 test trajectories and three resolutions. The primary post-hoc summary averages the four constant-shift residuals within seed at OOD-512, then averages seeds. The 98.75% crossed intervals adjust the four arm summaries. Paired hard-minus-free contrasts have descriptive 95% intervals and do not replace the frozen gauge test.

Table 11: Normalized whole-history constant-shift residual at OOD-512. Smaller values mean closer constant-shift equivariance, independently of exact output conservation.

Training	Normalized residual [98.75% CI]
Free absolute	0.0089 [0.0083, 0.0094]
Free residual	0.0083 [0.0081, 0.0086]
Hard absolute	0.0106 [0.0102, 0.0111]
Hard residual	0.0092 [0.0090, 0.0096]
Hard minus free	Difference [95% paired CI]
Absolute	0.0018 [0.0014, 0.0022]
Residual	0.0009 [0.0007, 0.0011]

Table 12: Advecting-sine residual at OOD-512, relative to the nonzero analytic conserving response. Intervals are descriptive 95% crossed bootstrap intervals.

Training	α/σ	Physical residual [95% CI]
Free absolute	-0.5	0.0144 [0.0136, 0.0153]
Free absolute	-0.2	0.0142 [0.0134, 0.0150]
Free absolute	+0.2	0.0142 [0.0134, 0.0152]
Free absolute	+0.5	0.0146 [0.0136, 0.0156]
Free residual	-0.5	0.0123 [0.0117, 0.0129]
Free residual	-0.2	0.0123 [0.0118, 0.0129]
Free residual	+0.2	0.0123 [0.0118, 0.0129]
Free residual	+0.5	0.0123 [0.0117, 0.0129]
Hard absolute	-0.5	0.0191 [0.0182, 0.0202]
Hard absolute	-0.2	0.0193 [0.0183, 0.0204]
Hard absolute	+0.2	0.0193 [0.0183, 0.0204]
Hard absolute	+0.5	0.0192 [0.0182, 0.0203]
Hard residual	-0.5	0.0138 [0.0131, 0.0145]
Hard residual	-0.2	0.0138 [0.0131, 0.0145]
Hard residual	+0.2	0.0138 [0.0131, 0.0145]
Hard residual	+0.5	0.0138 [0.0131, 0.0145]

A separate condition shifts only the last input frame. It is an input-sensitivity measurement, with no physically consistent history reference. At equal amplitude the ten-frame intervention has ten times its input energy, so their magnitude difference is not an equal-energy causal contrast. All signed amplitudes, resolutions, per-seed responses and marginal intervals are provided in the accompanying scientific JSON. Neither Advection symmetry is asserted for shallow-water depth.